PRIORITY QUEUES

Problem Solving with Computers-II





Emergency Room Triage

You're a software engineer at a hospital.
Patients arrive and need to be treated in order of severity.

1 = Critical (life-threatening)
2 = Urgent
3 = Moderate
4 = Minor

Emergency Room Triage

Priority Queue (ADT)!

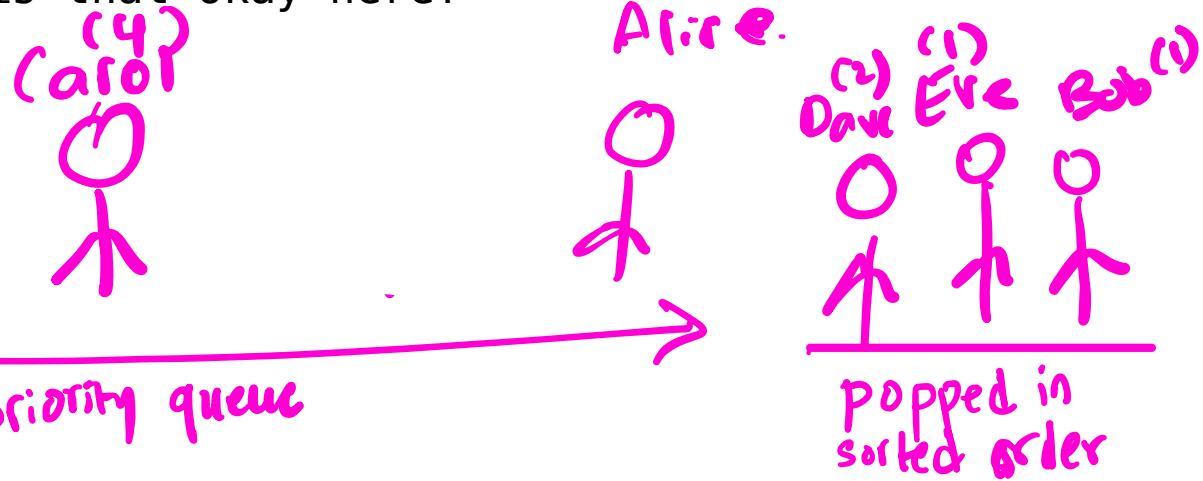
What if we used std::queue? X

```
    t=1  Alice   (Moderate)   ← treated first
    t=2  Bob     (Critical)   ← waits behind Alice
    t=3  Carol   (Minor)
    ...
```

First come, first served – is that okay here?

Whose next? Bob (1) — computed in O(1)

In a Priority Queue, the highest priority key can be looked up in O(1). "Highest priority" is always in the front.



C++ Priority Queue ADT

```
priority_queue<int> pq;

pq.push(20);
pq.push(20);
pq.push(80);
pq.push(50);
pq.push(100);

cout << pq.top();

pq.pop();
```

By default elements are compared using < operator

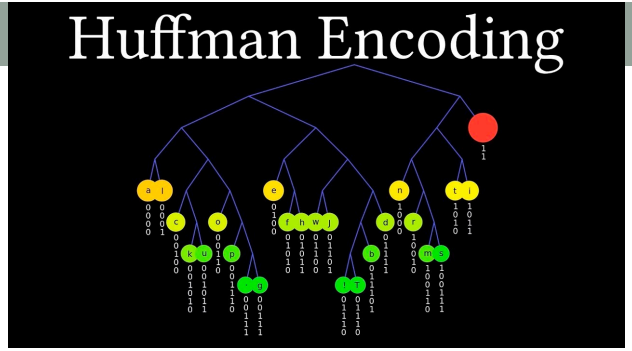
a < b means a has lower priority than b

// 100 → Largest element always on the Top

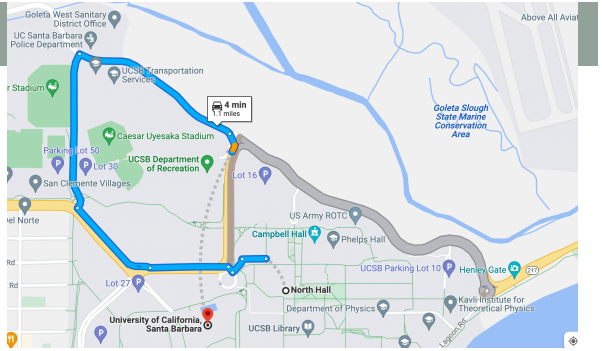
// highest priority (largest key) is deleted.

Don't call pop on an empty queue.

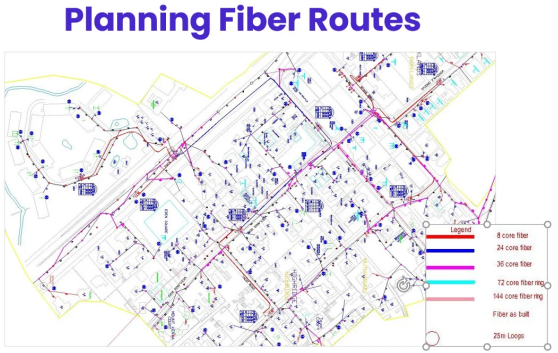
Live code ERTriage



Data Compression



Google Maps Navigation



Network Design

Algorithms: **Huffman Coding** \ **Shortest Path** | **Minimum Spanning Tree** /

ADT: **Priority Queue**

Data structure: **Binary Heap**

Complete Binary Tree

Vector

Many algorithms need to compute the min OR max repeatedly.

Priority Queue is used to speed up the running time!